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Transportation Planning Aspects of a Smart City- Case Study of GIFT City, Gujarat

D Srinivasa Reddy [a]*, Dr. K V Ganesh Babu [b], Dr. D L N Murthy[c]

[a,b&c] *Gujarat International Finance Tec - City Co. Ltd, Zonal Facility Centre, Block-12, GIFT CITY, Gandhinagar-382355, India.*

Abstract

Gujarat International Finance Tec-City Company Limited (GIFTCL) is developing a financial Central Business District (CBD) namely GIFT City between Ahmedabad and Gandhinagar as a global financial and IT services hub, a first of its kind in India. GIFT City is being developed as a high quality commercial zone along with an ideal blend of residential and social facilities that optimise land and real estate values with global connectivity and generation next infrastructure. GIFT City transport master plan aims at a transit oriented development based on walk to work approach, pedestrian friendly infrastructure with zero fatal accidents, easy and fast mobility with minimum conflicts and efficient public transport systems which reduce the per capita energy basket. In accordance with the GIFT City transport vision, the transport infrastructure has been planned to cater to the daily demand as well as that of the peak hour and Multi Level Parking (MLP) lots have been proposed at suitable locations in the GIFT city. To enhance the mobility and safety in GIFT city, various smart transport applications like advanced traveller information system, security and surveillance system, intelligent parking management system, parking guiding system, automatic vehicle location and tracking, real time travel response etc are planned. GIFT City transport master plan addresses the needs of all users of GIFT City, thereby it can be called as "User-Centred Smart Transport System" and its makes it possible for all GIFT City users to enjoy a world class life style in this vibrant global city. GIFT City smart transport planning parameters and design details are elaborated in this paper.

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Keywords: Smart city; Transit oriented development; Smart transport system; Public transport; Multi level parking.

* Corresponding author. Tel.: +91 79 3001 8437; fax: +91 79 3001 8321.
E-mail address: nivas.zeal@gmail.com

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1. Introduction

Gujarat International Finance Tec-City (GIFT), a Global Financial and IT/ITES Hub in the State of Gujarat, a first of its kind smart city in India, is being developed by Gujarat International Finance-Tec City Company Limited (GIFTCL), a Joint Venture of Gujarat Urban Development Company Ltd. (GUDCL) and Infrastructure Leasing & Financial Services (IL&FS). GIFT is strategically located between Ahmedabad and Gandhinagar at a distance of about 12 kms from Sardar Vallabhbhai Patel International Airport and 8 kms from Gandhinagar. The site is in close proximity to the National Highway (NH-48). The location of GIFT City is shown in Figure 1.

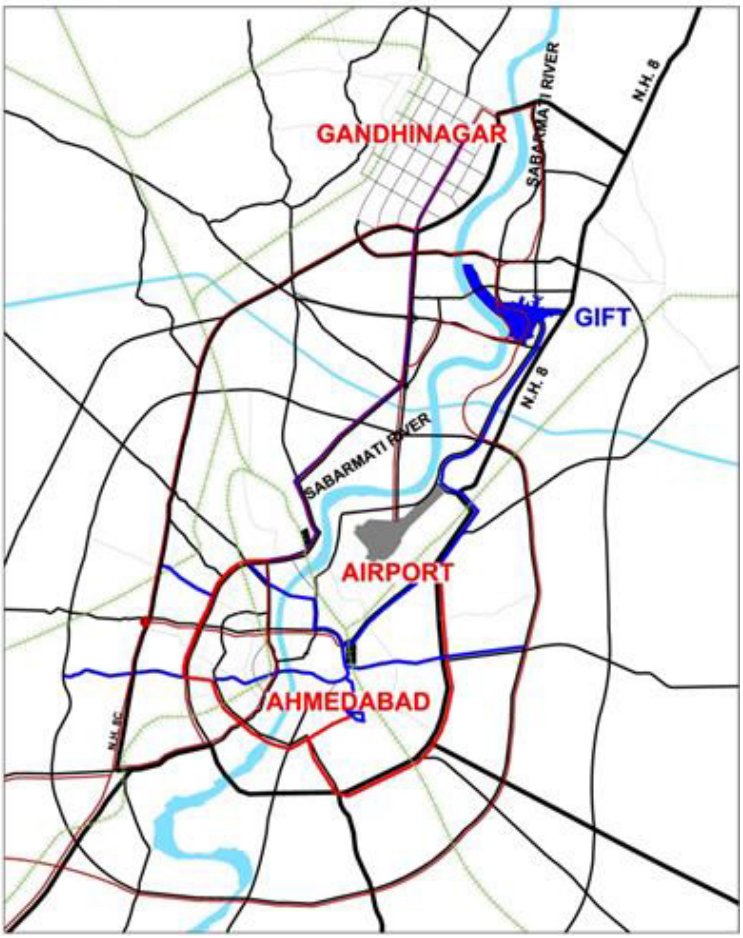

Figure 1: GIFT City location

GIFT City is being developed as a high quality commercial zone along with an ideal blend of residential and social facilities that optimise land and real estate values with global connectivity and generation next infrastructure. GIFT City is expected to be above or at par to all other international financial districts in terms of scale, scope, finest quality of infrastructure and life. GIFT City will encompass an area of 358 Ha (886 acres) with a total Built-up area ("BUA") of around 62 mn sft developed in three phases as Domestic Tariff Area (DTA) and Special Economic Zone (SEZ). The predominant land use is Commercial 67%; Residential 22% and Social 11%. The GIFT City master plan is presented in Figure 2.0

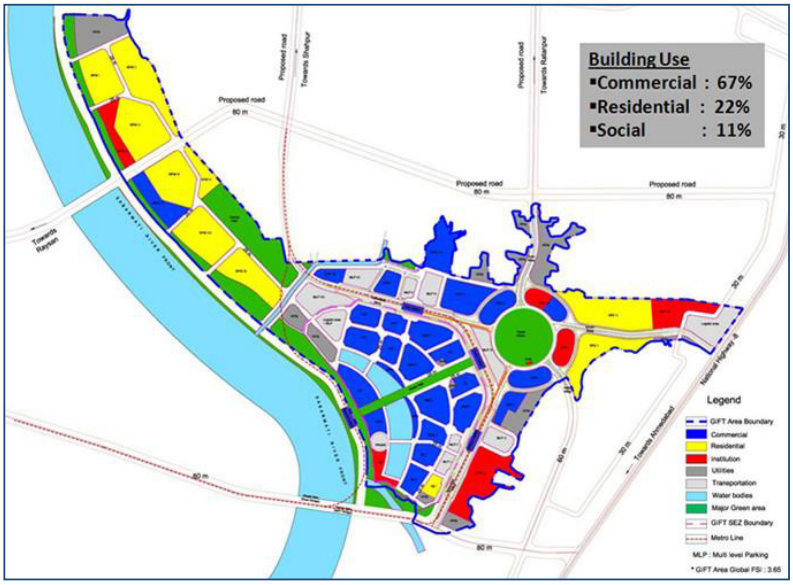

Figure 2: GIFT City master plan

2. Transport philosophy

GIFT City aims at a transit oriented development based on walk to work approach, pedestrian friendly infrastructure with zero fatal accidents, easy and fast mobility with minimum conflicts and efficient public transport systems which reduce the per capita energy basket. In accordance to the various activities envisaged for GIFT, the mandate is not only to provide connectivity, but also to cater to the daily demand as well as that of the peak hour with an objective to achieve higher share of public transport. The transport infrastructure in GIFT City has been planned to achieve this objective. Multi Level Parking (MLP) lots have been proposed at strategic locations in the GIFT City. To enhance the mobility in the GIFT City, various facilities such as underpasses, elevated covered walkways with travellators have been planned for pedestrian comfort and safety. Activity nodes have been integrated with the pedestrian network to enhance the walking experience. The public transport system is being integrated effectively connecting Ahmedabad and Gandhinagar.

3. Travel demand estimation

3.1 Daily trips

GIFT would act as a catalyst for development in its hinterland. The primary employment in GIFT is expected to give rise to equal number of secondary employment. The SEZ is expected to have an international clientele, and hence offices will be operating in multiple shifts. A second shift of work is considered in the processing area of the SEZ. 50% of the employment, primary and secondary, of the SEZ processing area is assumed to constitute in the second shift. The total number of trips generated due to office activity in the SEZ and non SEZ (DTA) areas are considered as incoming and outgoing trips.

3.2 Peak hour trips

Trips due to office use are considered in the peak hour as office is the predominant landuse of GIFT. Since there are two shifts, the peak hour occurs in the overlap between the general and the second shifts. This is the duration when the general shift is ending, and the second shift is about to start. The number of trips in the peak hour is considered as one-way trips of the primary jobs of the general and second shifts. Secondary jobs are considered to

occur in the off peak hours. It is easily appreciated that the peak direction of travel is that of the trips due to primary employment in general shift. Since there are a number of offices of varying kinds, it is assumed that the total number of trips due to primary employment in the general shift would occur in two hours. For sake of convenience, the number of trips in the peak period is spread equally over two hours.

3.3 Modal split

The vision of GIFT is to provide a robust and efficient public transport system in such a way that the modal split of the trips is 90:10 in favour of public transport. However, the present share of public transport in the modal split experienced in most Indian urban centres is in between 40-70%. Initiatives to boost public transit are undertaken nowadays. Appreciating the fact that it would take some time for people to shift from their personalized modes to public transport, a buffer is incorporated in the design of transport infrastructure. Keeping in view the 90:10 modal split, public transport is envisaged to cater to 90% of the trips. The share of private vehicles is designed as 25%, instead of 10%, so that a buffer is inbuilt into the system.

4. Public transport

GIFT would not only attract people from the nearby urban centres, Ahmedabad and Gandhinagar, but also encourage development in the nearby areas. Therefore, it is crucial to consider the development of the surrounding region of GIFT in design of transport infrastructure. Connectivity to the hinterland is considered while designing the public transport system for the city. Three corridors of Mass Rapid Transit (MRT) are planned to pass through the GIFT. These are essentially two MRT corridors, referred to as the West Corridor and the East Corridor. The West corridor branches out into two and rejoins near the north-west and south-west entry points. Thus, public transport connectivity is provided for the future realization of the skyline of tall buildings. The MRT stations (Refer Figure 3) are placed in maximum walking distance of 500m reach to the destinations. Bus Rapid Transit (BRT) system has also been proposed. Until MRT is operational, this would act as the medium of public transport connectivity. The bus stops are placed based on demand near major attraction points, such that they are located within 500m walking distance from the embarking/ disembarking points.

Figure 3: Proposed metro station in GIFT City

To achieve the envisaged public transport share of 90%, it is very essential to have a convenient and reliable public transportation system to GIFT City from the nascent stages itself. At present there is no public transport facility to GIFT City either from Ahmedabad or Gandhinagar and the proposed MRT & BRT operations to GIFT City may take some more time, considering GIFT City transport vision and realistic prevailing scenario, GIFT City started its own air-conditioned bus service from Ahmedabad (end point connected to the existitng BRT station at Visat, Ahmedabad) (Refer Figure 4).

Figure 4: GIFT City bus

5. External connectivity

The influence area of GIFT City includes Ahmedabad, Gandhinagar and the surrounding Districts of Kalol, Saiz, Sanand, Mehmedabad and Dehgam. GIFT City will act as a catalyst for all round development in the entire region.

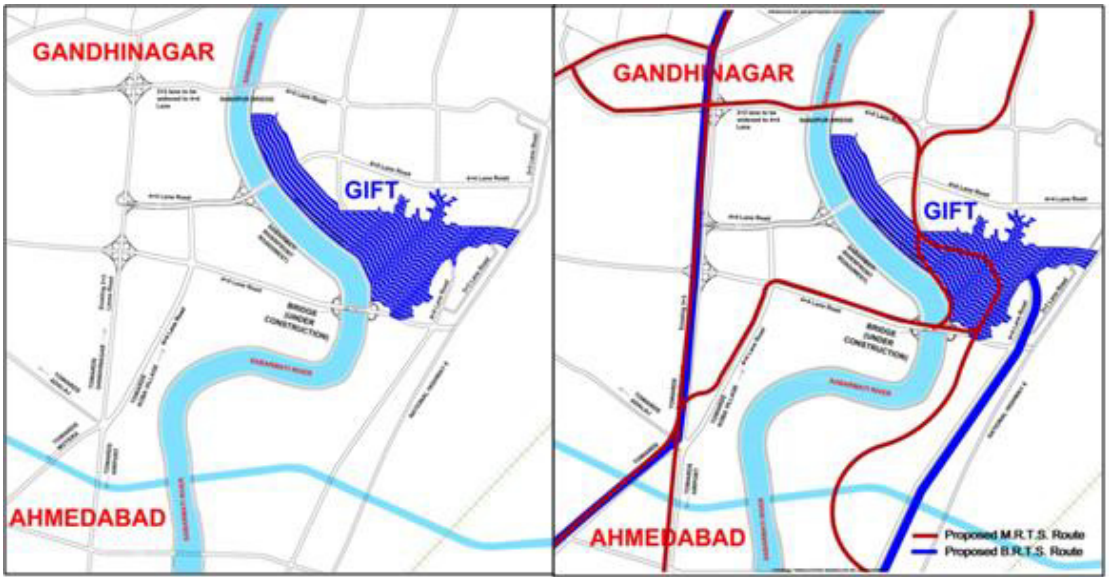

Figure 5: GIFT City external connectivity

The major road network in the influence area includes National Highway 48 (NH-48) on the East, Ahmedabad-Gandhinagar Highway on the west passing through PDPU circle and Koba circle. Connectivity is proposed from all the sides in the form of bridges and major road corridors, apart from these, river front roads are proposed on either side of the Sabarmati River and the same are presented in Figure 5.

6. Internal transport

The principle behind design of movement within the GIFT City is to have dedicated corridors for vehicular

movement, and movement in the internal areas would be mainly pedestrian with a provision for movement of emergency vehicles when required. There are five major entry points to GIFT Project, connected to the regional road network. The arterial transport corridor comprises of dual 4-lane main arterial road with dual 2-lane sub-arterial roads on either side. The five roads converge into a central roundabout, which is designed to effectively handle the total traffic volume entering from all directions. The radius of the rotary is designed at 200m. The length of the shortest section of the roundabout is 150m. Six lanes have been provided for carrying mixed traffic which enables to handle 8000 PCU/hour.

The other components of internal transport infrastructure include flyovers and underpasses for maintaining a conflict free movement within the GIFT City. Large green corridor is designed within the rotary consisting of 32 acre area for various social activities includes golf course, club etc. For the movement of pedestrian's between the central roundabout and the adjoining packages, pedestrian underpasses are planned (Refer Figure 6). In the SEZ area, the network would be mainly for pedestrian movement and interspersed nodes that connect to the buildings in the enclave with elevated walkway and travelator connectivity.

Figure 6: GIFT City internal Roads

7. Pedestrian facilities

Pedestrian walkways will be developed keeping in mind pedestrian safety and comfort. Street furniture will be included and activity centres introduced at nodes for making walking a safe and comfortable experience. Shady trees and comfortable ergonomic seating arrangements will be made. Level differences will be minimized for comfort of the elderly.

Figure 7: Walkability influence area around MRT station

The material suggested for pedestrian pathways is grass pavers which are not only comfortable to walk on, but also strong enough to carry the load of emergency vehicles such as fire tenders. In course of time when subsequent phases of development come into place, and demand of pedestrian infrastructure increases and hence, it has been envisaged to construct elevated walkways from major transport nodes such as MRT stations and multi level car parks for direct connection into buildings. These walkways will be covered and could even be air conditioned for complete pedestrian comfort. Walkability influence area around each MRT station is presented in Figure 7.

8. Streetscape

As GIFT City is proposed to be highly dense with commercial development, in the planning stage itself, sufficient buffer is left for making provision of taxi bay, bus stop, free flow of pedestrian movement including drop off and pick up from the respective buildings. By considering these, buffer 22 m is left from the edge of the carriageway to the building footprint. This area will also facilitates to lay the various utility infrastructures such as water, irrigation, automated waste collection, power, ICT corridors (Refer Figure 8).

Figure 8: Streetscape section as planned (L) and implemented (R)

9. Signage system

As GIFT City aims at high share of public transport, transit oriented development (walk to work approach) and pedestrianised oriented internal movement etc., the need for good signage to make these activities easier becomes more and more apparent. Considering these aspirations GIFT City developed GIFT City Signage System to "provide a clear visual language and graphic standards that can be universally understood, encourage walking and transit usage by offering quality multi-modal information, and provide consistent information across a broad range of environments in the city at par with best national and international standards." Armed with a novel design approach, which stressed on a seamless integration of information and user experience, the project focused on factors like progressive disclosure, predictability, importance of landmarks, human ergonomics – all lending it a unique touch. The system gives the information in a progressive manner, not just letting the user know the orientation through maps and distance with respect to time, but also with an indication, a physical reference, a pre-view to where the user’s next step can lead, or simply put, a nudge further in the right direction. As a part of phase 1, few sign posts as per GIFT City signage system are erected on site and the same are presented in Figure 9.0

Figure 9.0 : Installed monolith & finger post sign boards

10. Smart transportation system

GIFT City, proposed smart transportation system improve transport outcomes such as transport safety, transport productivity, travel reliability, informed travel choices, environmental performance and network operation resilience. Due to the proposed development in GIFT City, the interest in smart transport comes from the problems caused by traffic congestion and a synergy of new information technology for simulation, real-time control, and communications networks.

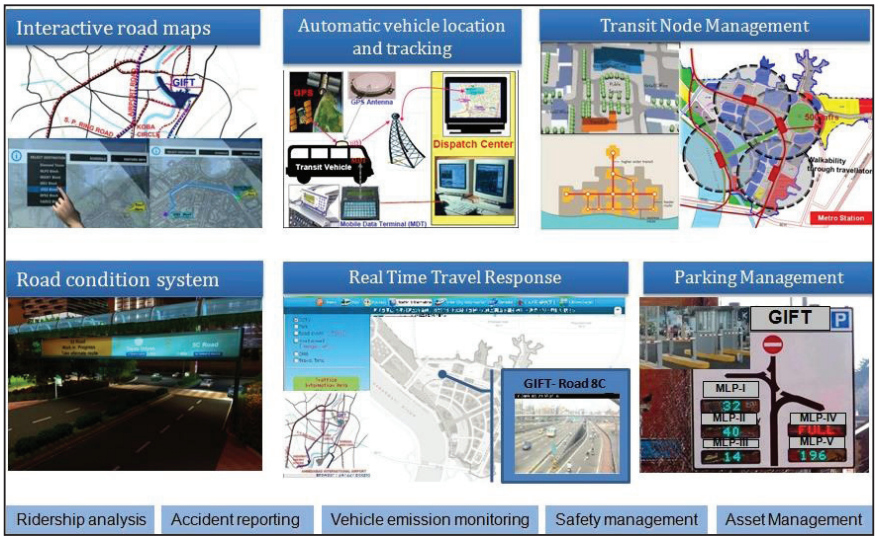

Figure 10: GIFT City smart transportation

From smart transport consideration it would include advanced traveller information system for communicating the information through variable message signs etc; security and surveillance system; intelligent parking management system to operating the parking areas; advanced public transport system to operate the MRT and BRT by sharing the information with the end users. The gambit of smart transport also includes automatic vehicle location and tracking, real time travel response etc. All the required provisions to facilitate smart transport applications in GIFT City are considered in the planning stage and itself and incorporated in the design stage for successful implementation of smart transport system in the GIFT City. Illustrative example of proposed GIFT City smart transport system is shown in figure 10. As a part of phase wise smart transportation implementation GIFT City installed GPRS based real time vehicle trackers in all GIFT City vehicles. The installation of these trackers resulted in better planning and fleet management and helping in speed checks of vehicles, accurate location, deviations in the journey paths(if any) etc. Glimpse of one such vehicle tracking application implemented in public transport is presented in figure 11.

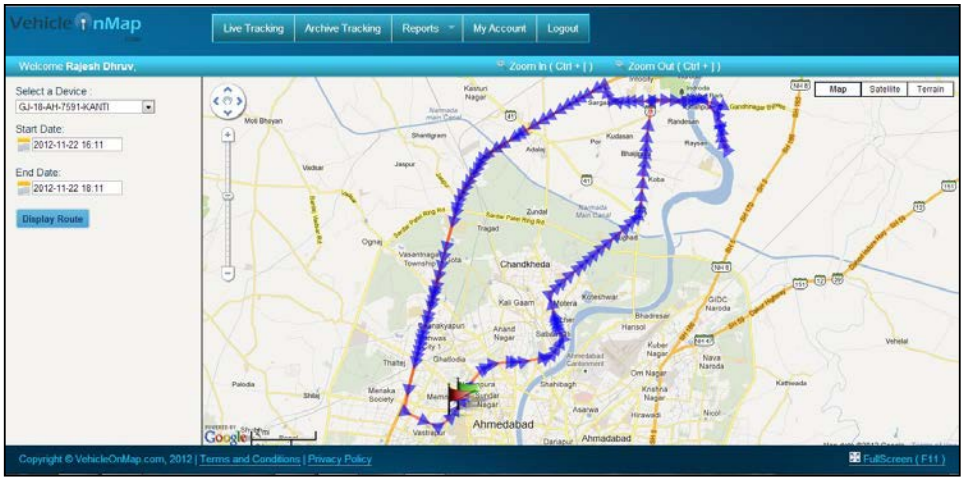

Figure 11: Vehicle tracking application

11. Parking philosophy

One of the most commonly faced problem in urban metros is parking. GIFT City has incorporated sufficient provisions in the GIFT City master plan and its development control regulations (DCR) so that there won't be any parking problems in the GIFT City.

Figure 12: MLP Locations (L) and proposed MLP-2 architectural view (R)

As a part of GIFT City master plan, MLPs have been proposed at strategic locations (refer figure 12) so that there won't be any parking issues. As per the GIFT City DCR, parking requirement for different land uses are estimated by considering the parking norms. In addition to this, 10% for visitor and 20% for two wheeler parking need to be provided. The total requirement of parking as per DCR for all phases is approximately 1,50,000 Equivalent Car Spaces (ECS) out of which approximately 65,000 ECS need to accommodate in different common parking areas designated in the Master Plan.

11.1 Multi level parking

As mentioned in the previous sections, GIFT City project is developing in a phased manner so development of required transport infrastructure also is being developed in the same manner. As a part of phase1 infrastructure development GIFT City is developing an MLP with a capacity of approximately 5400 Equivalent Car Spaces (ECS). Parking Guidance System (PGS) and Parking Management Systems (PMS) are also part of the proposed MLP development.

11.1.1 Parking guidance systems

Parking guidance systems basically, to assist drivers in efficiently locating available parking spaces, and hence minimising risk of associated delays within the car park and on the upstream road network. PGS shall monitor the parking spaces in real-time using ultrasonic sensors. Driveway sensors shall be installed in the main entrance to count the vehicle in real-time until the time it gets a parking. The sensor shall be ultra slim so that it will not affect the height limitations of the parking & Ultra flat no angle loosing visibility LED indications at entrances, ramps, crossings & parking places and shall provide a far view of clear parking spaces at least 60 meters. Visibility of LED lights can be adjusted thru the software installed on PC. The proposed PGS, system configuration and real time application images are shown in figure 13.

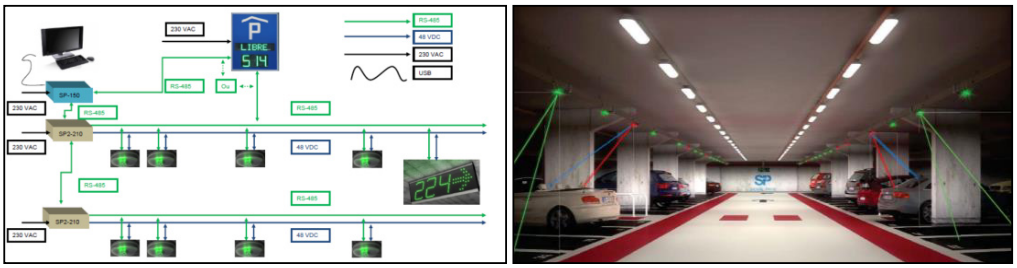

Figure 13: PGS- system configuration (L) real time application (R)

11.1.2 Parking management system

Parking Management System (PMS), mainly consists of access and revenue controls to assist operators in efficiently managing entry, exit and payment for use of the car park facility, by minimising both customer inconvenience and cost of fare recovery. Parking Management System comprises of gate entry barriers, long range RFID reader, entry station, exit station, cash terminal, handheld unit, pay on foot systems, central management station etc. The glimpses of the proposed units are presented in figures 14 & 15.

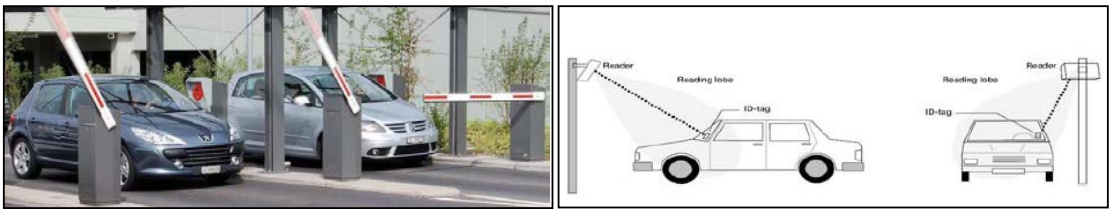

Figure 14: PMS- access control systems (L) & long range RFID reader (R)

Figure 15: PMS-revenue control systems (pay on foot (L),smart card reader (R))

12. Conclusions

GIFT City is based on Transit Oriented Development approach where metro stations are strategically located at the center of high-density commercial development with progressively lower-density development spreading outward from the center.

GIFT City transport plan envisages a modal split of 10:90 between private and public transport. However the system has been designed as 25%, private instead of 10%, so that a buffer is inbuilt into the system. To meet the envisaged public transport share and considering the prevailing conditions GIFT started its own public transport facility.

Its proposed to develop MLPs with total capacity of 50,000 ECS in GIFT. People coming by private vehicles can walk to their destinations after leaving their vehicles in these MLPs which are located strategically near the entry points on either side of the road.

Proposed smart transportation applications in GIFT City improve transport outcomes such as transport safety, transport productivity, travel reliability, informed travel choices, environmental performance and network operation resilience

GIFT City transport master plan addresses the needs of all users of GIFT City, thereby it can be called as "User-Centred Smart Transport System" and its makes it possible for all GIFT City users to enjoy a world class life style in this vibrant Global City.

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